DAV CSS 1022 MINI PROJECT
NETFLIX MOVIES AND SHOWS ANALYSIS REPORT

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Course: CSS 1022 Data Visualization

1. Introduction and Objective of the Analysis:

The entertainment industry has completely changed the way of consuming entertainment, with Netflix emerging as a leading platform worldwide.

This analysis uses a Netflix dataset to explore patterns in its content library, focusing on key variables: **content type, genre, country, release year, and duration.**

The objective is to understand how Netflix's catalog is and how it adapts to audience preferences and global trends. The study analyses content distribution, growth patterns, and audience interests which can help content creators, marketers, and platform strategists improve content planning, target audiences effectively, and enhance global reach in the competitive streaming market.

2. Data Understanding and Exploration

Source: Kaggle - Netflix Movies and TV Shows dataset.

Records and Attributes

The dataset contains information about Netflix movies and TV shows, including attributes such as type, title, director, cast, country, release year, rating, duration,

genre, and description. Each record represents a unique title identified by show_id.

Important Variables

The key variables used in the analysis: type, rating, country, listed_in (genre), and release_year

They help us understand the content distribution and trends. Additional variables like director, cast, and duration are used for deeper insights and conclusions.

3. Data Cleaning and Preprocessing

Need for Pre-Processing:

Data pre-processing is required to handle missing values, duplicates, and inconsistencies in the dataset. Cleaning and standardizing the data ensures accurate analysis and prevents wrong results. Good data quality improves the reliability of the conclusions and supports better decision-making.

Steps Taken in Data Cleaning

- Checked dataset shape
- Identified missing values and duplicates - isnull().sum() and duplicated().sum()
- Filled missing categorical values
- Removed missing rows - dropna()
- Cleaned text data - str.strip()
- Eliminated duplicates. - drop_duplicates()
- Verified the cleaned dataset
- Generated summary statistics. - describe(include='all')
- Identifying outliers through box plot

String manipulation & aggregation techniques:

- groupby() and value_counts() were used for aggregation
- str.strip() and str.split() were used for string manipulation

Cleaning:

```python
# -------- BEFORE CLEANING --------
print("Shape BEFORE:", df.shape) #- shape: returns number of rows and columns

print("\nMissing values BEFORE:\n")
print(df.isnull().sum()) #- isnull(): checks missing values

print("\nDuplicate rows BEFORE:", df.duplicated().sum()) #- duplicated(): finds duplicate rows
```

```
Shape BEFORE: (7973, 12)

Missing values BEFORE:

S               0
type            0
title           0
director        0
cast            0
country         0
date_added      9
release_year    0
rating          0
duration        3
listed_in       0
description     0
dtype: int64

Duplicate rows BEFORE: 0
```

```python
# -------- CLEANING --------

# Fill missing categorical values
df['director'] = df['director'].fillna('Unknown') #- fillna(): fills missing values
df['cast'] = df['cast'].fillna('Unknown')

# Drop rows with important missing values
df = df.dropna(subset=['country', 'rating']).copy() #- dropna(): removes missing rows

# Clean string columns (remove extra spaces)
df['country'] = df['country'].str.strip() #- strip(): removes extra spaces
df['rating'] = df['rating'].str.strip()
df['type'] = df['type'].str.strip()

# Remove duplicates
df = df.drop_duplicates()  #- drop_duplicates(): removes duplicates


# -------- AFTER CLEANING --------
print("\nShape AFTER:", df.shape)

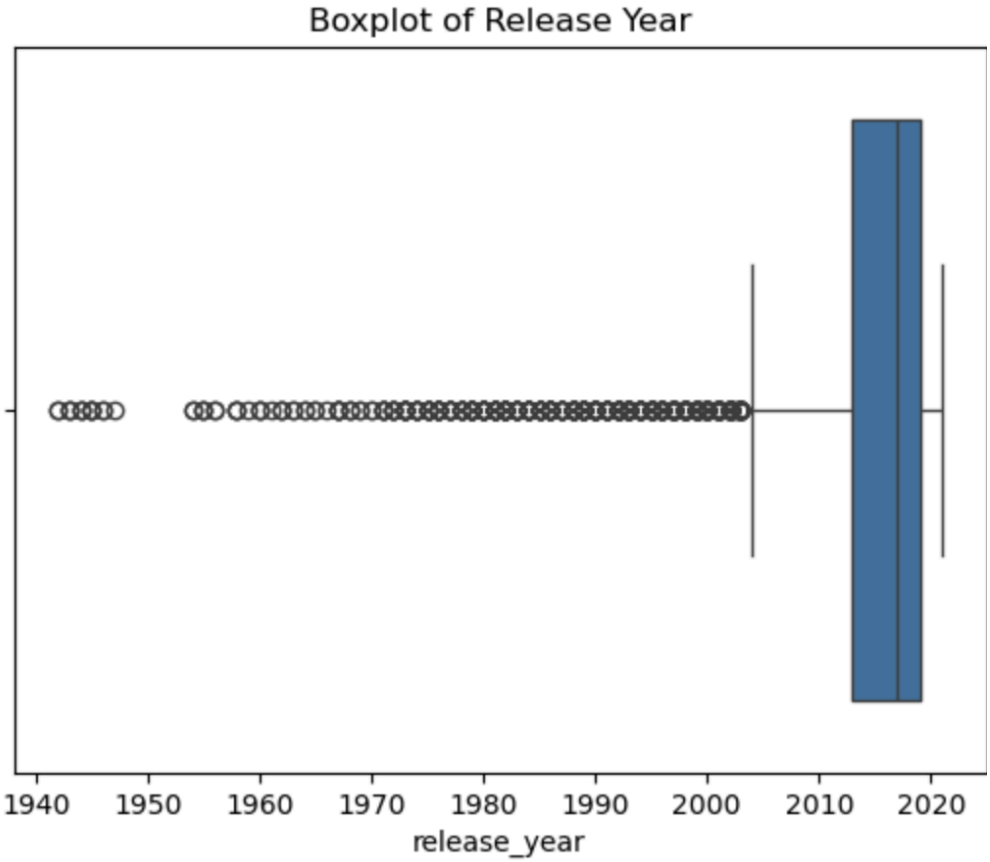
print("\nMissing values AFTER:\n")
print(df.isnull().sum())

print("\nDuplicate rows AFTER:", df.duplicated().sum())

print("\nSummary AFTER:\n")
print(df.describe(include='all'))  #- describe(): gives summary statistics
```

4. Visualization and Analysis

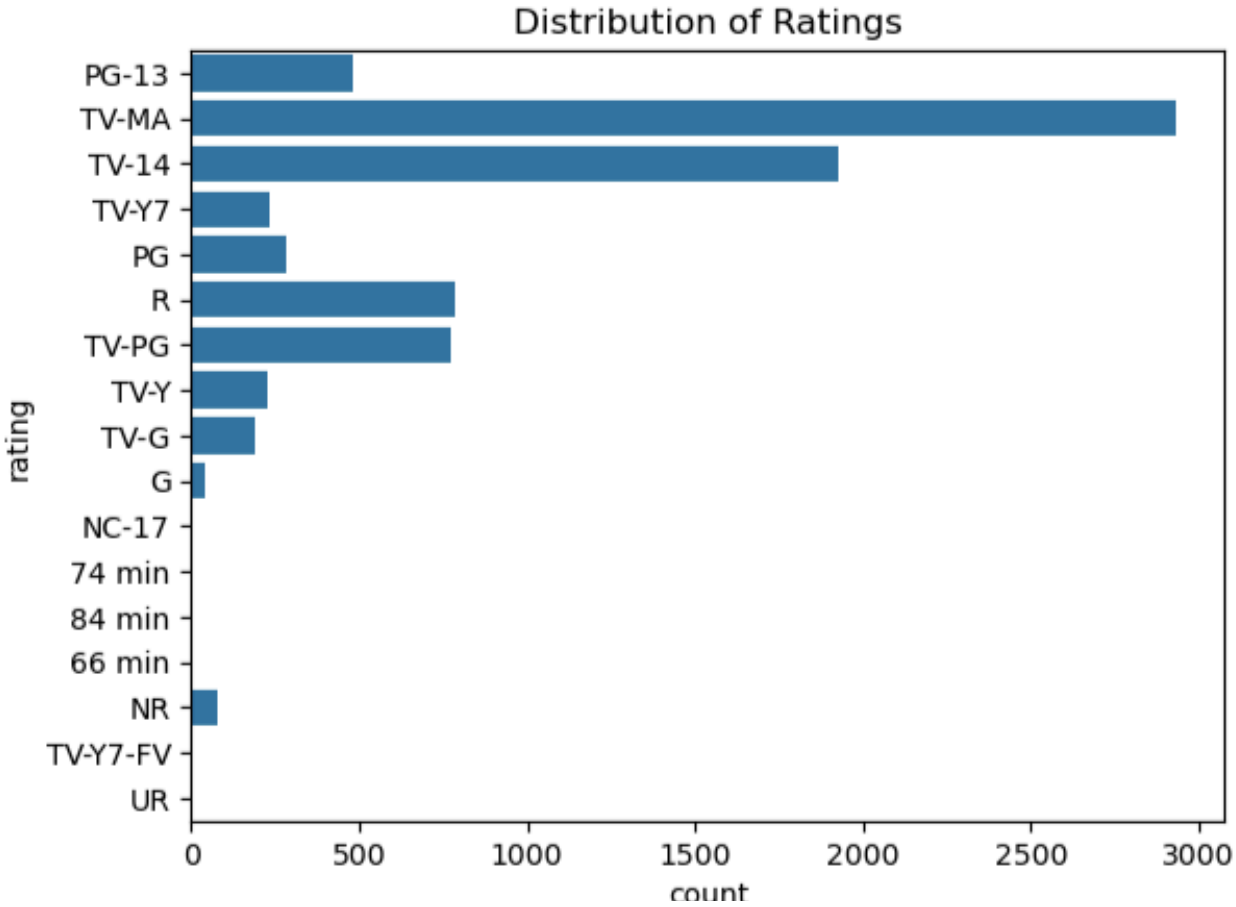
1. Identify outliers in release year - boxplot



Observations:

- A large number of older release years appear as outliers on the left side (denoted by the dots), indicating that few of the titles belong to earlier decades.
- The wide spread of outliers highlights a long historical range of content, but they are much lesser than the content released in recent years.

2. Distribution of Ratings: - Countplot

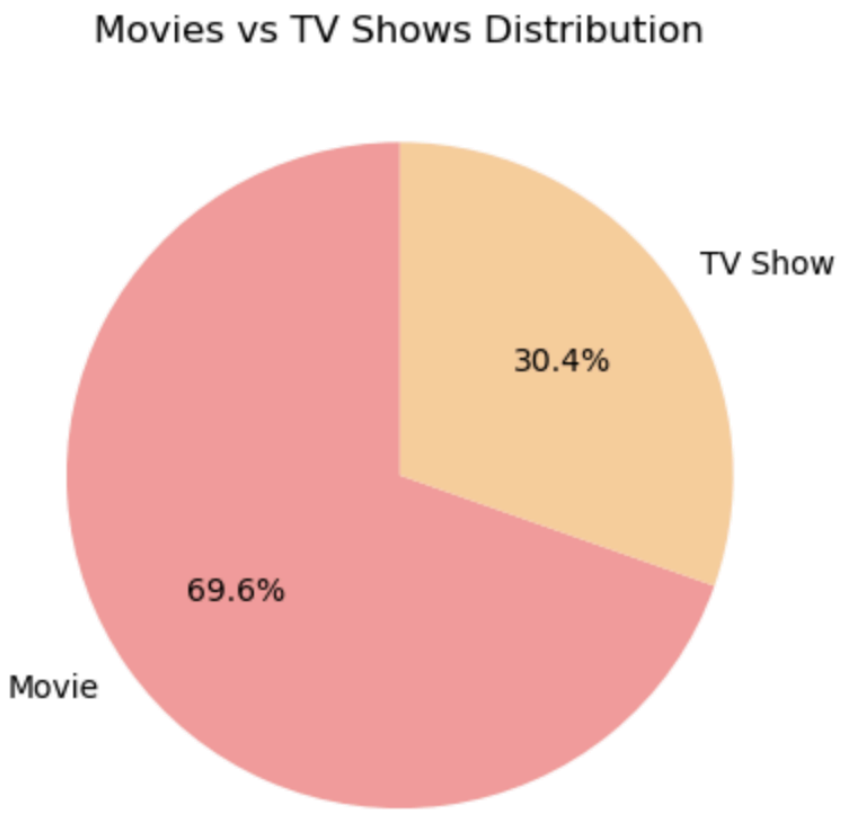


Observations:

- The graph shows that “TV-MA” is the most common rating on Netflix, followed by TV-14, indicating that the platform mainly focuses on mature and teen audiences.
- A few incorrect values like “74 min” appear, showing data inconsistency in the rating column which is an error during the making of the dataset.

3. Movies vs TV Shows

Graph type: Pie Chart -

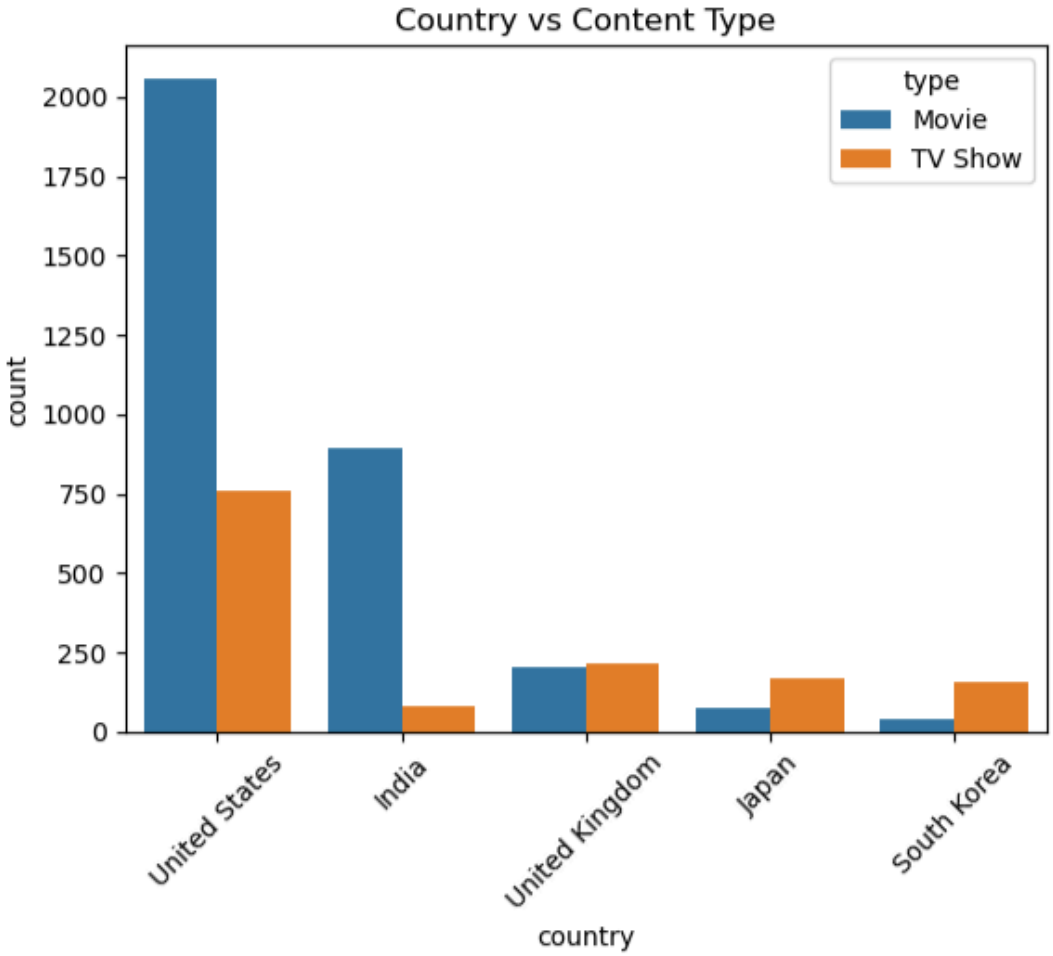


Observations:

- The chart shows that Movies make up a larger portion of Netflix content compared to TV Shows, indicating that the platform focuses more on movie content.
- TV Shows, while fewer, still play a key role in user retention, since series keep viewers coming back for multiple episodes. They also ensure the constant usage of Netflix as an OTT Platform.

4. Country vs Content Type

Graph Type: **Count Plot**

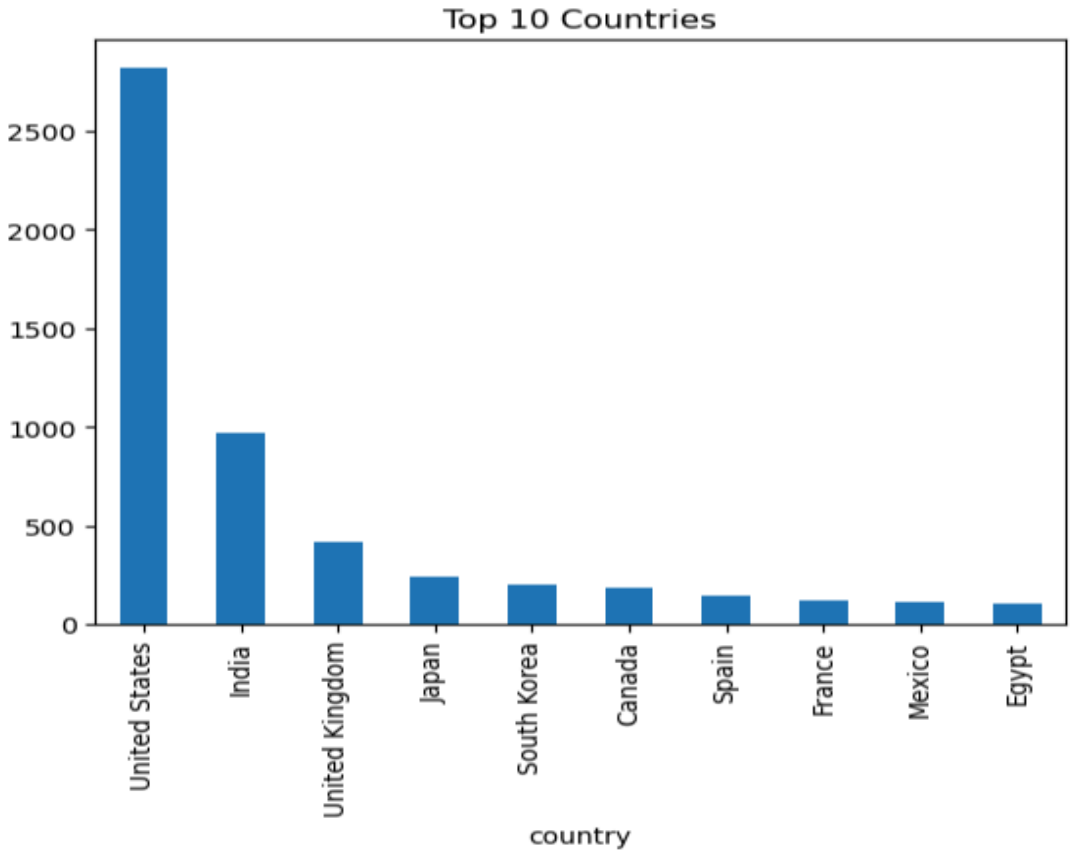


Observations:

- The graph shows that the United States produces the highest number of both Movies and TV Shows, followed by India.
- Movies dominate in most countries, while some countries like the United Kingdom and Japan show a relatively balanced distribution between Movies and TV Shows.
- This shows that Netflix content production is heavily concentrated in a few countries, with varying preferences for Movies and TV Shows.

5. Top 10 Countries producing Netflix content:

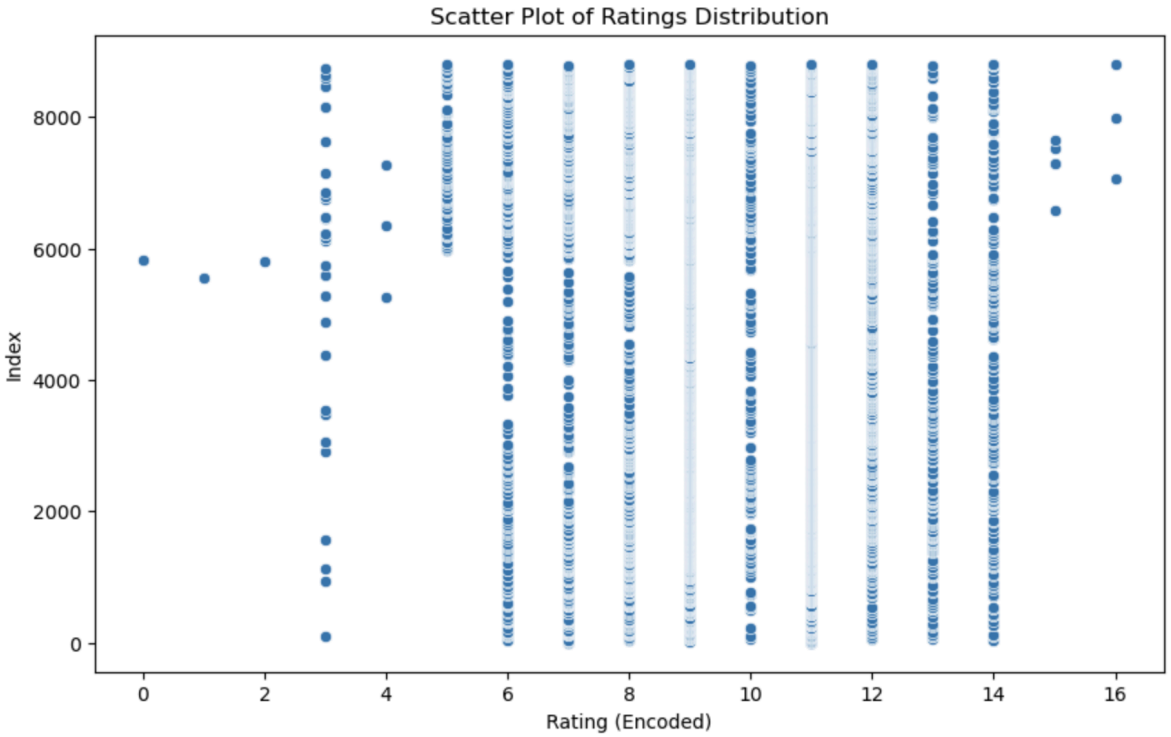
Graph Type: **Bar plot**



Observations:

- The graph shows that the United States produces the highest amount of Netflix content by a large margin, followed by India and the United Kingdom.
- This indicates that content production is highly concentrated in a few countries, with the remaining countries contributing comparatively less.
- There is a significant gap between the top country and others, showing the dominance of the United States in Netflix content production.

6. Visualize distribution of content ratings using a scatter plot

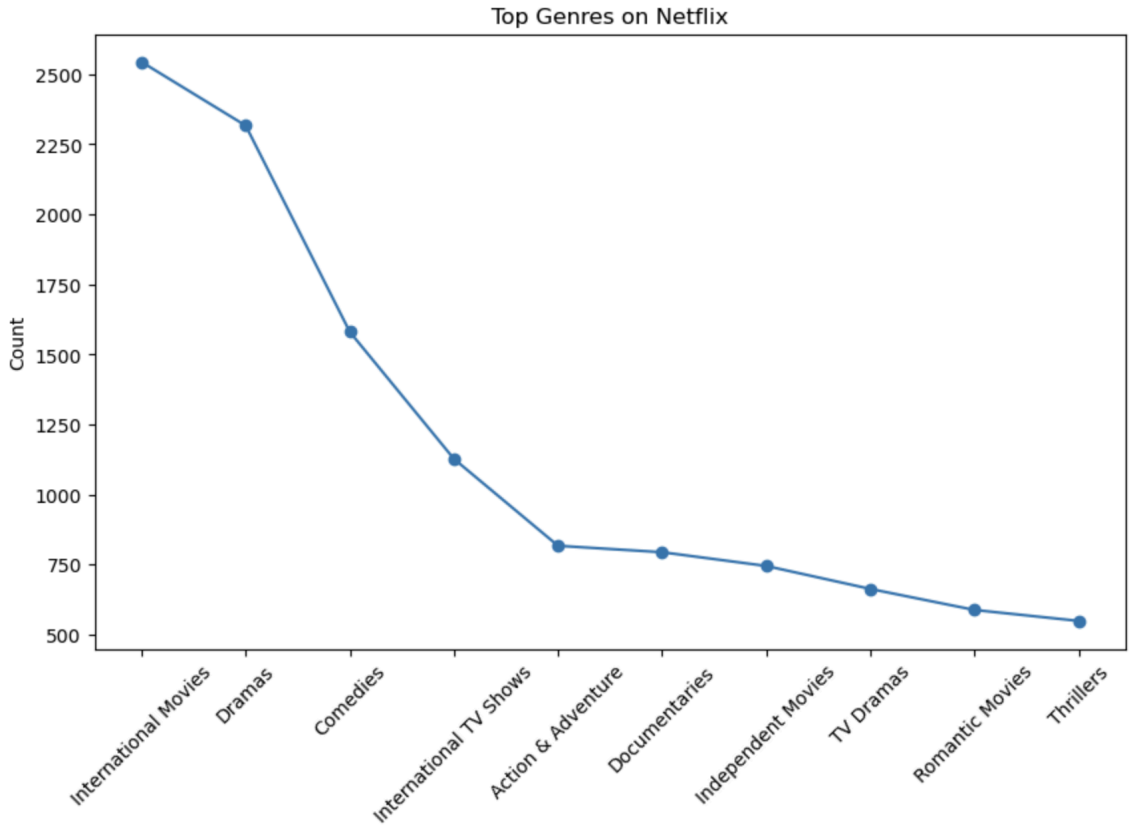


Observations:

- The scatter plot shows vertical clusters of points for each rating category, which shows that many data entries share the same ratings.
- Certain rating categories have more density of points as per the graph, which shows that those ratings are more common.
- The absence of any clear pattern or trend indicates that ratings are categorical in nature and the data is not continuous.
- This confirms that scatter plot is not ideal for categorical data like in this dataset.

7. Visualize the variation of top genres

Graph**: Line Plot**

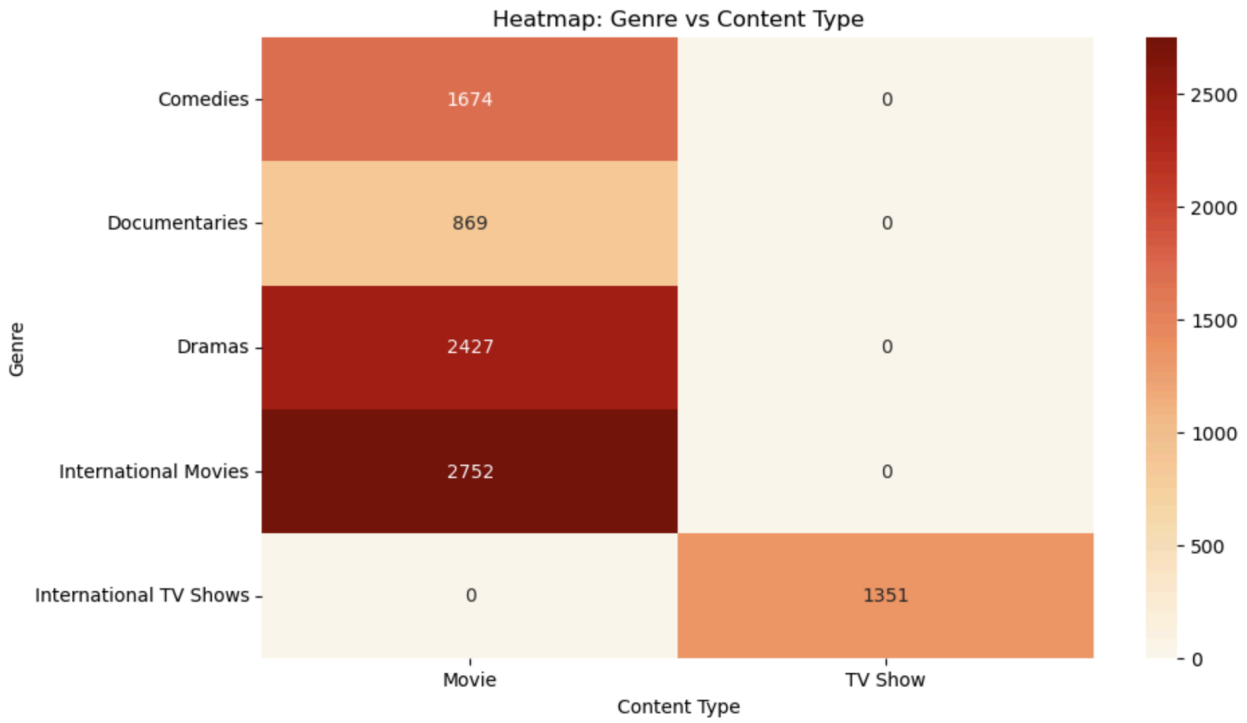


Observations:

- The plot shows that 'International Movies' is the most dominant genre, followed by 'Dramas', showing a strong preference for these categories on Netflix.
- There is a sharp decline in counts after the top two genres, showing that only a few genres contribute significantly.
- Genres like Comedies and International TV Shows have moderate presence, while others such as Thrillers and Romantic Movies appear less frequently.
- The downward trend in the line indicates an uneven distribution of genres, with content heavily dominated by certain genres.

8. Cross Tabulation visualised by heatmap

Graph Type: **Heatmap**

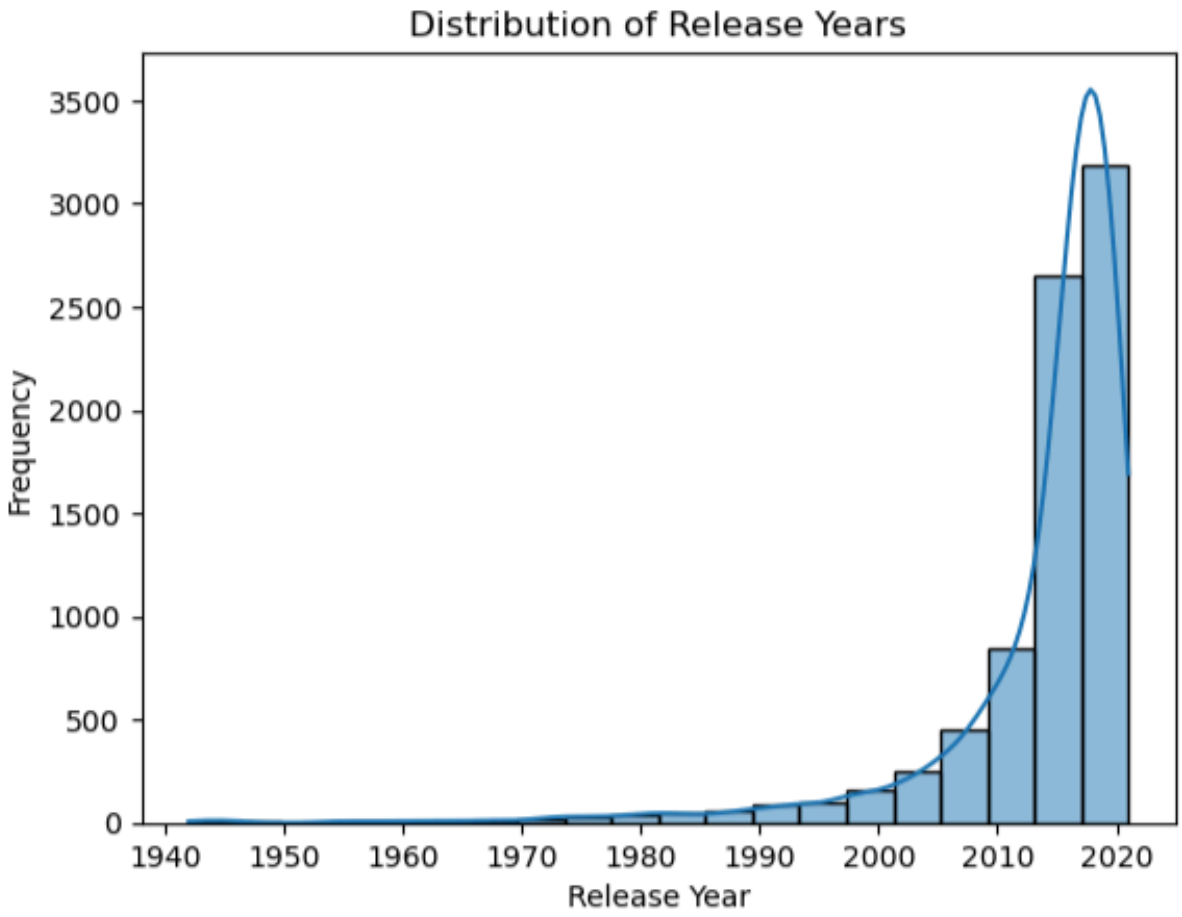


Observations:

- The heatmap shows that certain genres like Dramas and International Movies are more common in Movies compared to TV Shows.
- Darker shades indicate higher frequency, revealing which genres dominate each content type. Few genres have no TV Shows present in Netflix.
- The presence of multiple content types within the same entry highlights the diversity of Netflix content.

9. Distribution of Release years:

Graph Type : **Histogram** with **KDE**

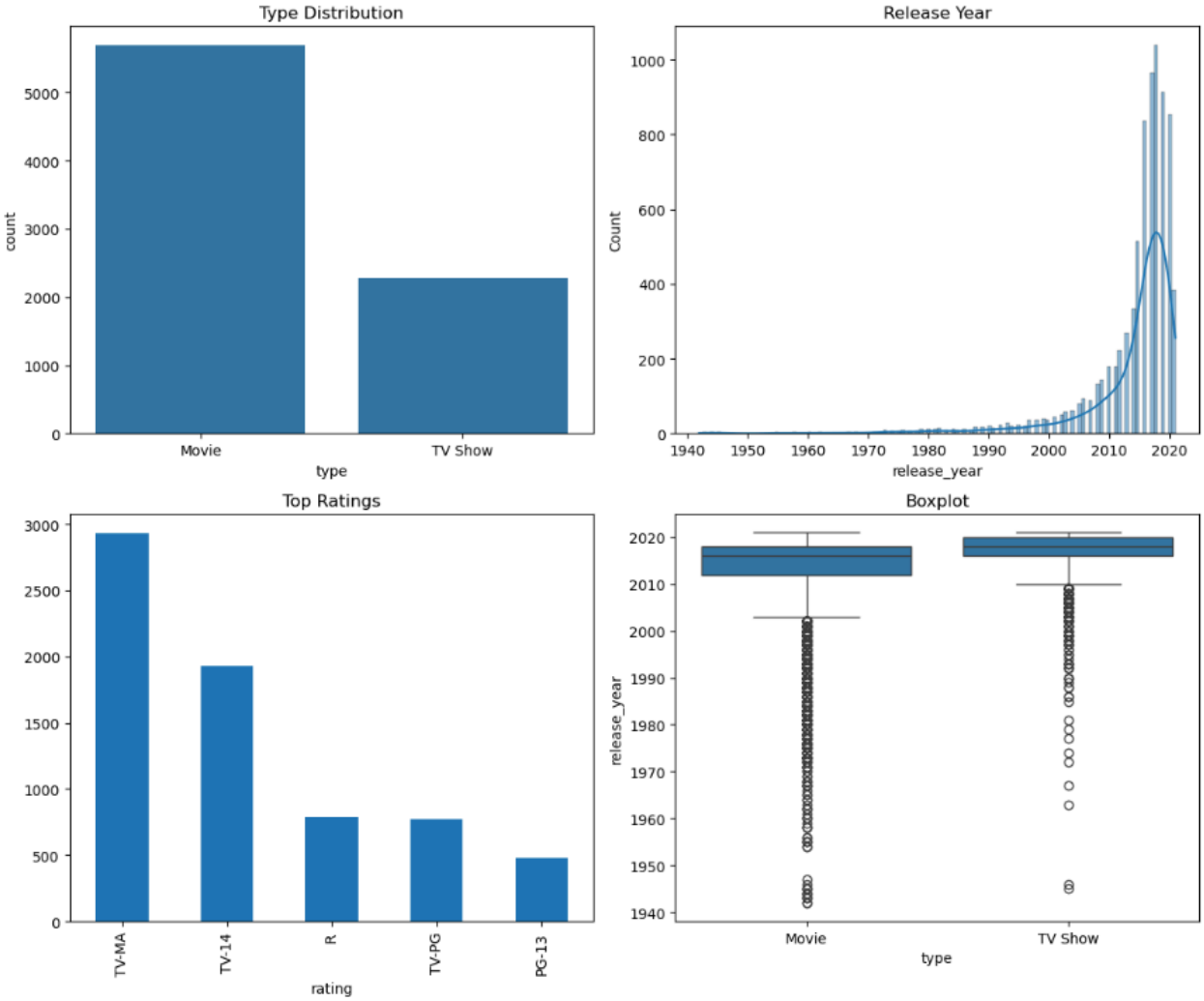


Observations:

- The graph shows that most Netflix content is concentrated in recent years, especially after 2010, indicating a rapid growth in content production.
- The KDE curve highlights a peak in recent years, while older years have significantly fewer releases.
- The right-skewed distribution indicates that Netflix has significantly expanded its content library in the last decade.

10. Easier Understanding through Multi plot

Graph Type: **Multi Plot**



Observations:

- The combined plots provide an overall understanding of the dataset.
- The countplot shows that Movies dominate over TV Shows, the histogram indicates that most content is released in recent years, the bar chart highlights the most common ratings, and the boxplot reveals that older content appears as outliers.

5. Key Insights:

- The dataset has categorical data. Movies dominate the Netflix platform compared to TV Shows, indicating a stronger focus on films.
- Netflix strategically targets adult audiences aged 17+, likely due to higher engagement and financial profits.
- A few genres like International Movies and Dramas contribute the majority of content, while other genres have lower representation.
- Content production is mostly by a few countries, especially the United States and India, with significantly higher contributions than others.
- Scatter and categorical analyses show no strong relationship between variables like content type and rating, indicating similar distribution patterns.

6. Tools Used:

- Python (Jupyter Notebook)
- Pandas and Numpy
- Matplotlib and Seaborn

7. Conclusion:

The analysis reveals that Netflix's content library is dominated by movies, with a strong focus on mature audiences and popular genres such as dramas and international content, which keeps the profits still high. Content production is largely concentrated in a few countries, highlighting uneven global distribution. Overall, the insights show that Netflix prioritizes widely appealing content while maintaining variety across genres and regions, supporting its global growth and improved audience engagement.